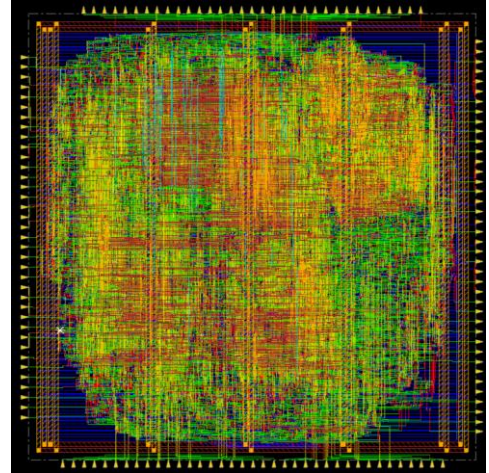


PPA Optimized 32-bit Pipelined ALU Design (RTL-to-GDS) (main version)



to



- **Target Frequency:** 1.66 GHz @ 45nm CMOS
 - **Timing Closure:** +21 ps Positive Slack
- **Power Consumption:** 53mW @ 1.66GHz
 - **Total Area:** 52000 μm^2 @ 1.66GHz
- **Verification:** 100% Toggle & Block Coverage
- **Implementation:** Full RTL-to-GDSII Sign-off Flow

This project is done for the ELE417 term project

The Cadence™ Tools used for this project is under educational license

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Design Goals:

In an era of rapid semiconductor evolution—where advanced chiplet architectures define the 2025 "State of the Art"—this project explores the boundaries of high-performance design using a foundational technology. While the **45nm Bulk CMOS** process utilized here represents the 2005–2011 industry era rather than today's sub-5nm nodes, it serves as a robust platform for demonstrating sophisticated architectural and physical optimization techniques.

- **Frequency Maximization:** Ramping performance to achieve a **1.66 GHz** operating frequency, demonstrating high-speed pipelining on a mature node.
- **Intelligent Power Management:** Implementing distinct power modes to save energy, with architectural hooks prepared for future **Power-Gating** integration during the physical hardening phase.
- **Area-Efficiency:** Minimizing the silicon footprint to reduce manufacturing costs, ensuring the design remains competitive in terms of Performance-per-Area.

By treating a mature process with modern optimization strategies, this project showcases the ability to extract maximum value and performance from established semiconductor technology.

Design Specifications:

1. Architectural Details

- **Pipeline Depth:** Two pipelined modules: ADDER32 & MULT32
- **Latency vs. Throughput:** Pipeline depths are 1 clock for ADDER32 and 7 clock cycles for MULT32 respectively. (latency) and how many results it produces per cycle (throughput).
- **LowPower mode:** My ALU has a mode called LowPower this ensures much lower power consumption compared to pipelined high throughput Turbo mode.
- **High-Frequency:** My desing can operate safely upto 1.66GHz at Turbo mode and can operate safely upto 180MHz at LowPower mode.

2. Physical & Environmental Specs

- **Technology Node:** 45nm Planar CMOS (GPDK45).
- **Voltage:** For high-frequency operation i needed a bigger overdrive voltage so my VDD: 1.2V But external smart controller can lower the voltage to 1.0V if operating frequencies are not near the upper limit. (1.66GHZ@Turbo & 180MHz@LowPower)
- **Corner Cases:** Mention which PVT (Process, Voltage, Temperature) corners you used for synthesis (e.g., "Slow-Max" for timing, "Fast-Min" for power).-----

3. Verification Metrics

- Logical correctness: %100 tested & verified (edge cases included)
- Behavioural correctness: %100 tested & verified (edge cases included)

Turbo mode specific:

- Pipeline conflicts:
- %100 tested
- If different blocks, make valid output at the same time we raise conflict_valid signal and Y_conflict is captured by external control_unit

Turbo mode <-> LowPower mode transitions:

- %100 tested.
- Turbo -> LowPower: No data loss, pipelined Turbo module finishes its current job and then turns off.
- LowPower -> Turbo: No data loss, pipelined Turbo modules power and pipeline counter starts from 0 so no unexpected output is generated.

Idle <-> ANY mode transitions:

- %100 tested.
- No data loss
- If control unit wants ALU to go into sleep -> ALU acknowledges this and finishes its current process at hand before sleep and takes no input unless idle is pulled low.
- If control unit wants ALU to go into LowPower mode and lowers its core frequency as well as voltage -> ALU acknowledges this and finishes its current pipelined process at hand before going into combinational LowPower mode.

4. Advanced Timing Data & performance

- **Setup/Hold Slack:** +5ps wns at hold time, +21ps wns at setup time
- **Frequency:** 1.66GHZ
- **Clock Uncertainty:** I budgeted a 50ps of very generous jitter while doing tests.
- **I/O Delay:** I budgeted 80-100ps of worst case I/O delay doing tests.
- Hence the architecture is pipeline throughput is 1.66 GOPs.

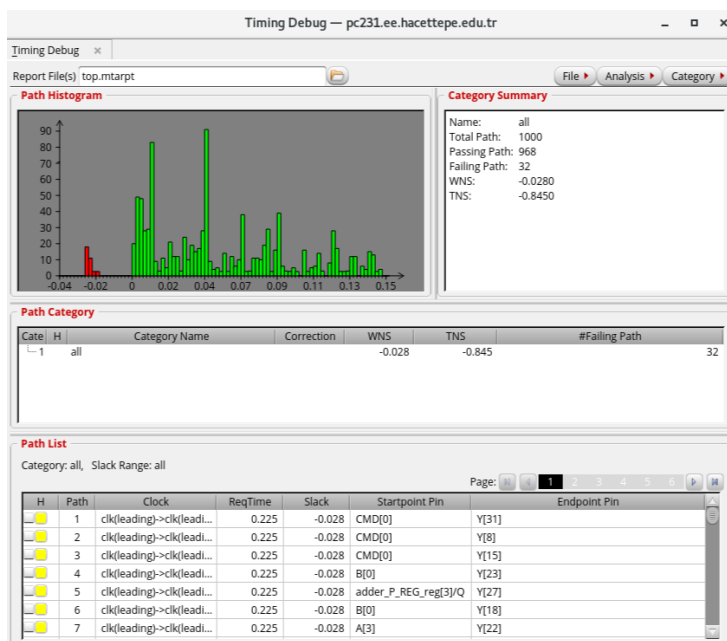


figure 1: Timing histogram generated by Genus for 2.816GHz

5. Power

This power usage is outputted from Genus. Hence the real power usage is expected to be %20-40 higher due to extra capacitances & buffers

Total Power (Turbo)	51.94 mW	@ 1.66 GHz, 1.2V (High Performance)
Total Power (Low)	3.30 mW	@ 180 MHz, 1.0V (Power Saving)
Power Density (Turbo)	100 W/cm²	High thermal density due to frequency/voltage
Power Density (Low)	6.35 W/cm²	Reduced density in power-saving mode

6. Instructions

- I thought of a very customized way of handling opcode. And used a 5bit opcode for mode selection inside ALU.
- The available instructions for my ALU are:

Opcode	Operation	Instruction	Description
00000	A+B	ADD	Addition
00001	A-B	SUB	Subtraction
00010	A>B	GT	Greater than
00011	A<B	LT	Less than
00100	A+B+C	ADDC	Addition with carry
00101	A-B-C	SUBC	Subtraction with borrow
00110	A-B	SUB	Subtraction
00111	A-B	SUB	Subtraction
01xxx	A*B	MUL	Multiplication
10000	A<<B	SLL	Shift Left Logical
10001	A<<<B	SLA	Shift Left Arithmetic
10010	A>>B	SRL	Shift Right Logical
10011	A>>>B	SRA	Shift Right Arithmetic
10100	A rot B	ROR	Rotate Right
10101	A rot B	ROL	Rotate Left
10110	Byte Swap	BYT	Byte swap

11000	$\sim B$	NOT	Bitwise NOT
11001	$A \oplus B$	XOR	Bitwise XOR
11010	$A B$	OR	Bitwise OR
11011	$A \& B$	AND	Bitwise AND
11100	$\sim(A \oplus B)$	XNOR	Bitwise XNOR
11101	$\sim(A B)$	NOR	Bitwise NOR
11110	$\sim(A \& B)$	NAND	Bitwise NAND
11111	$A == B$	EQ	Equality check

Total of 22 very fundamental instructions are implemented.

RTL Architecture:

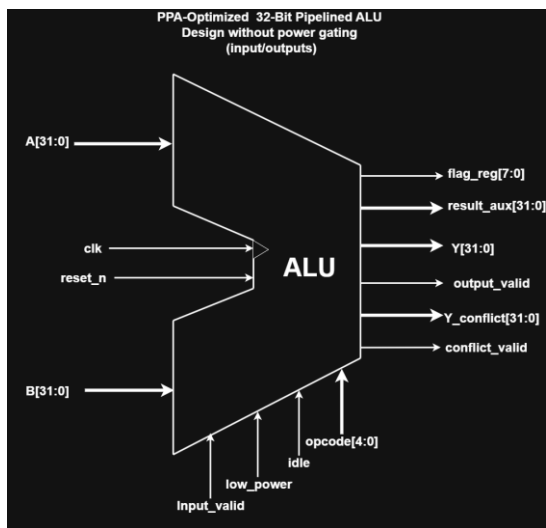


figure 3: I/O Level Block Diagram

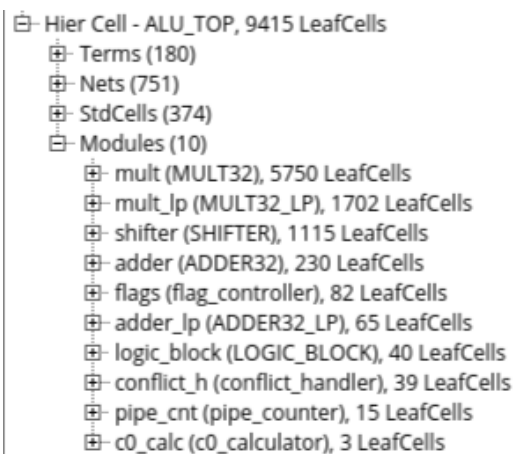


figure 4: Hierarchy of design

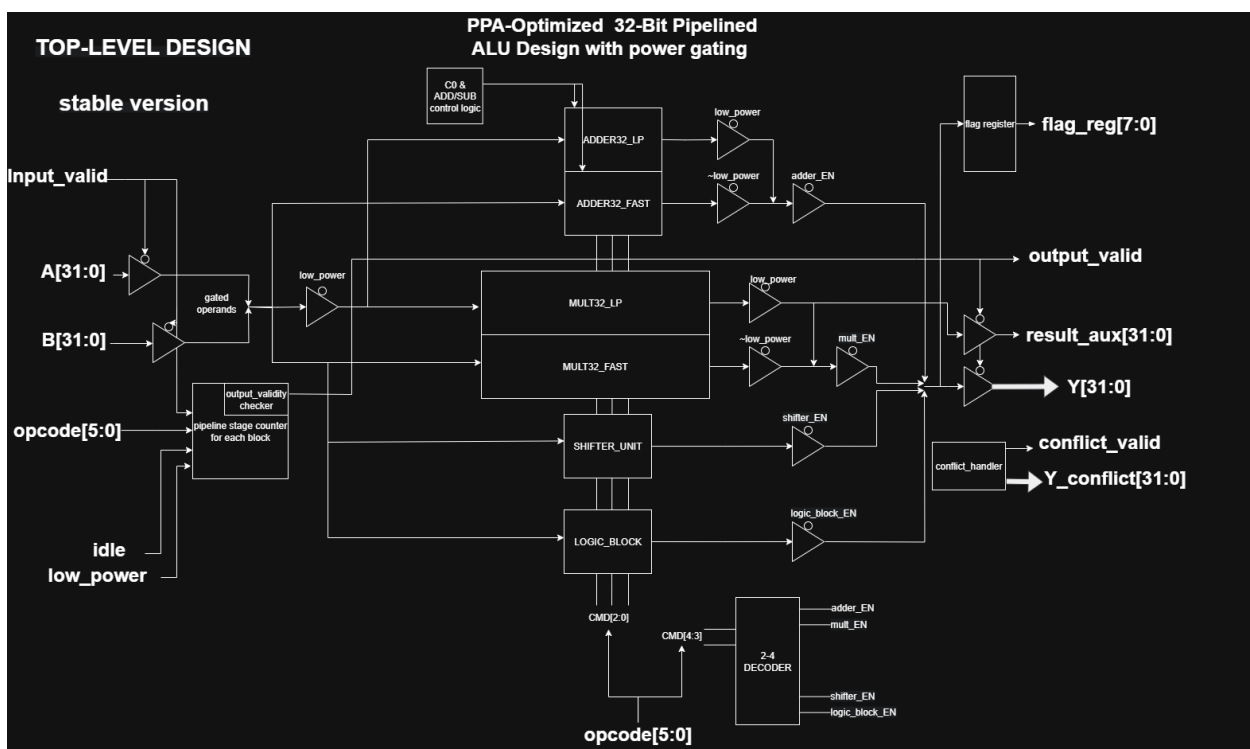


figure 5: Detailed architecture

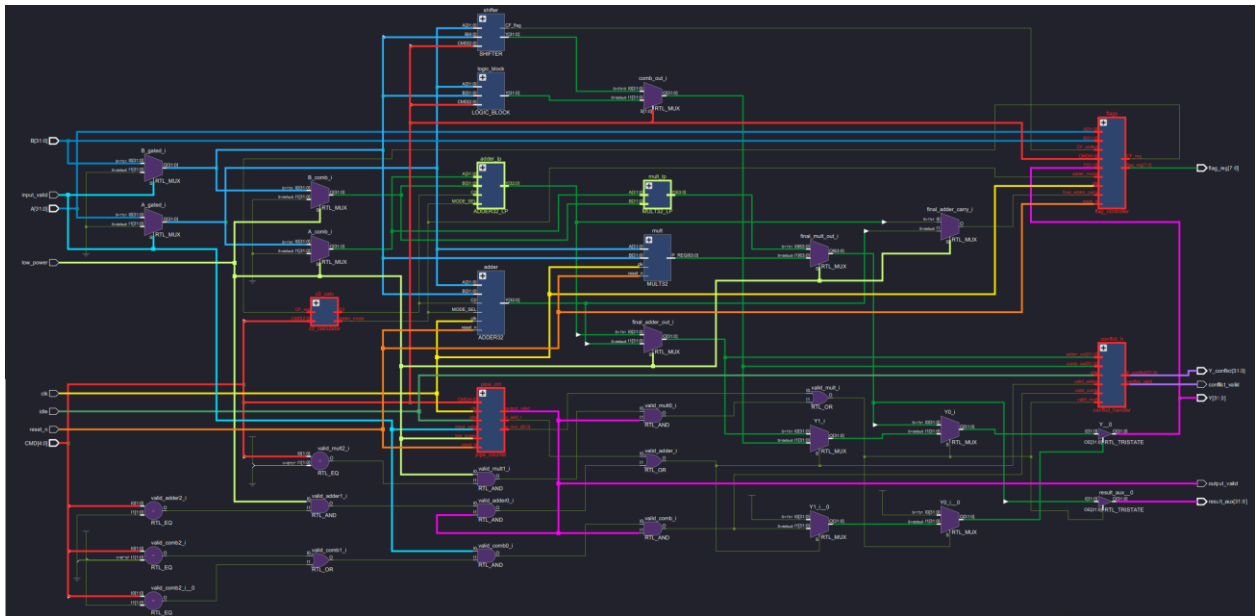
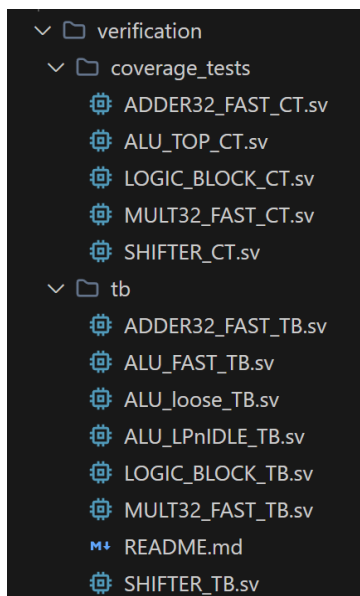


figure 6: Detailed full architecture

Design Flow:

Verification and Testing (Xcelium & Vivado):



I Self-Checking Environment: Integrated a **Golden Model Scoreboard** for automated, real-time comparison between RTL outputs and reference behavioral models.

Sign-off Coverage: Achieved **100% Toggle and Block coverage**, verifying that every logic path and bit-transition in the 32-bit datapath was fully exercised.

Edge Case Analysis: Identified two specific operand corner cases; these are managed via **Compiler-level scheduling** or **Control Unit bypass logic** to maintain stability without sacrificing the 1.66 GHz performance target.

figure 7: verification for coverage setup

Synthesis Results (Genus):

- Total cells: 10496

Netlist Statistics — pc231.ee.hacettepe.edu.tr			
Generated by: Genus(TM) Synthesis Solution 21.18-s082_1 (Jul 18 2023 13:08:41)			
Generated on: Dec 28 2025 18:29:31			
Module: design:ALU_TOP			
Technology libraries: fast_vdd1v2 1.0			
fast_vdd1v2 1.0			
Operating conditions: PVT_1P32V_0C (balanced_tree)			
Wireload mode: enclosed			
Type	Instance	Area	Area_%
Tristate	64	218.880	0.819
Sequential	1436	8864.640	33.154
Logic	8451	17179.994	64.253
Inverter	502	381.330	1.426
Buffer	43	93.024	0.348
Total	10496	26737.868	100

figure 8: Comparing cell types by area utilization.

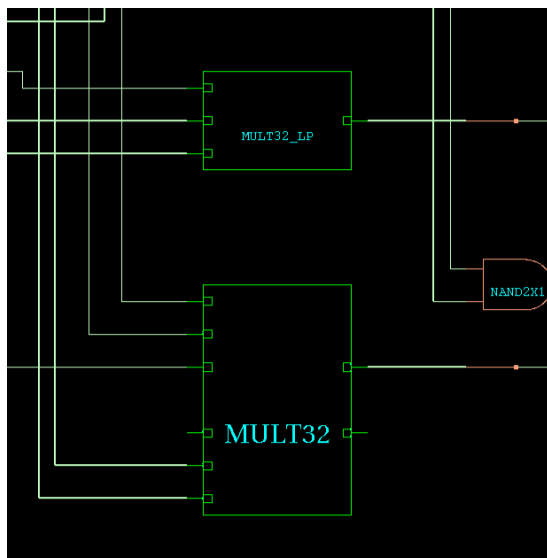


figure 9: Comparison of _LP & HP modules

Physical Implementation (Innovus):

- Floorplaning -> power routing -> pin/cell placing -> routing -> optimization -> CTS -> optimizations -> enabling onChipVaritions -> final setup&hold optimizations

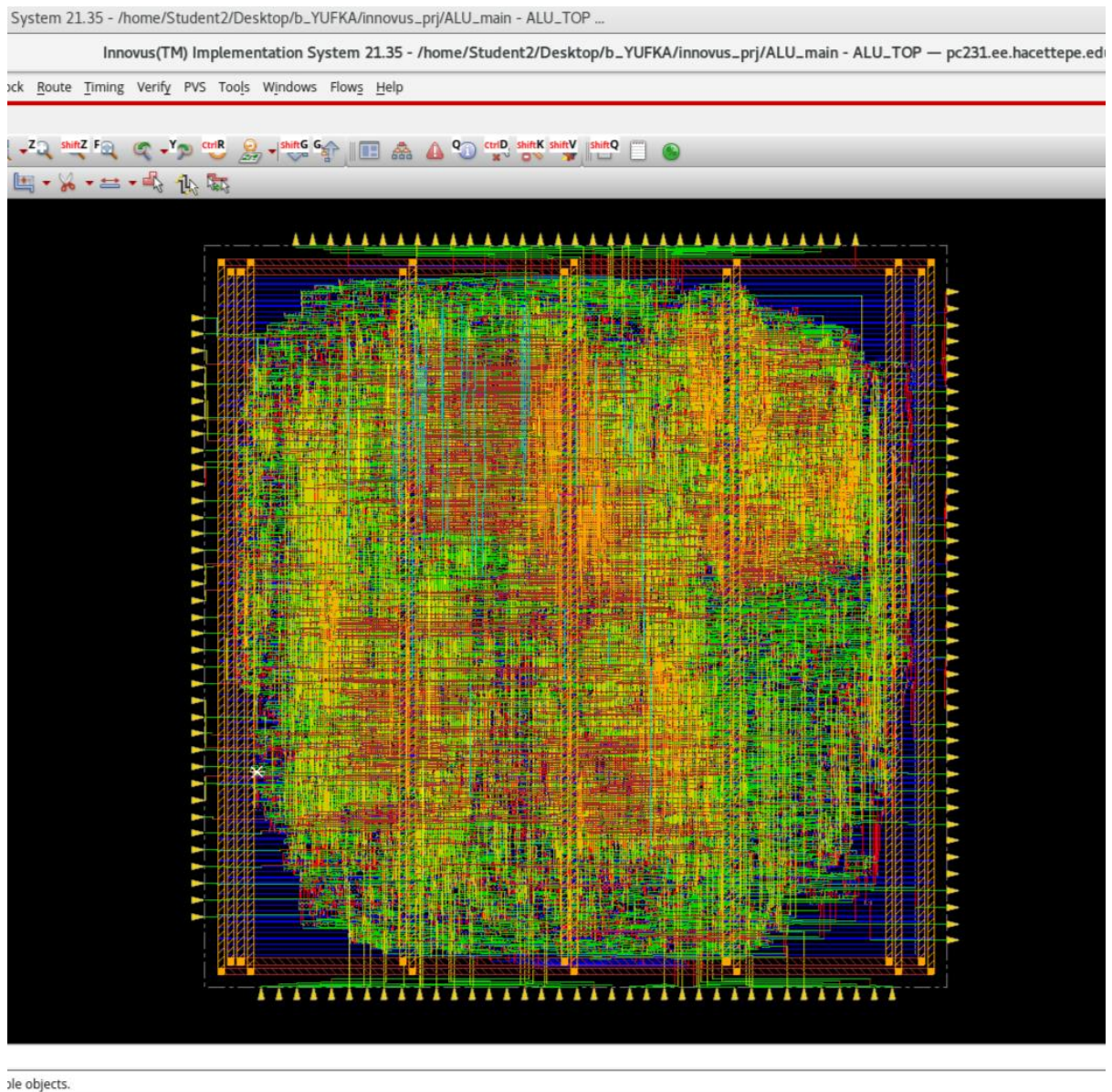


figure 10: Final layout output

Conclusion

This project successfully demonstrates a full RTL-to-GDSII flow for a 32-bit pipelined ALU. By achieving 1.66 GHz at 77.36% utilization with 100% functional coverage, the design balances aggressive performance targets with physical feasibility. The documented trade-offs between Turbo (51.94 mW) and Low Power (3.3 mW) modes highlight a robust architecture ready for integration into high-performance processor cores.

Future Work & Industry Outlook

- **Scaling to Advanced Nodes:** I am eager to transition this architecture to **FinFET** and sub-7nm nodes to manage complex effects like multi-patterning and interconnect resistance.
- **Sign-off Maturity:** I plan to advance the high-frequency and power-gated variants to a full **Pre-Sign-off** level, closing timing across all PVT corners.
- **Tool Appreciation:** The intelligence of **Cadence Genus and Innovus** in navigating thousands of PPA constraints has been a highlight of this project.
- **Career Goal:** This experience has solidified my passion for the full **Digital-to-Silicon** cycle and developed my skills for modern semiconductor engineering.

[Full Source Code & Documentation](#)

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